MILITARY SPECIFICATION

CLOTH, SATEEN, COTTON, FIRE-RETARDANT TREATED

This specification has been approved by the Naval
Air Systems Command, Department of the Navy.

1. SCOPE

1.1 This specification covers the requirements for one type of sateen cloth treated for flame resistance and used as black-out curtains in aircraft.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-P-1134	Packaging and Packing of Cotton and Cotton-Synthetic Fiber Blend Fabrics (Excluding Duck Fabrics)

STANDARDS

Federal

FED-STD-4	Glossary of Fabric Imperfections
FED-STD-191	Textile Test Methods

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes

(When requesting specifications, standards, and publications, refer to both title and number. Copies of this specification and applicable documents may be obtained upon application to the Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120. All other documents should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 8305

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2.2 Other publications - The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Rules and Regulations under the Textile Fiber Products Identification Act

(Application for copies should be addressed to the Federal Trade Commission, Washington, D. C. 20580.)

3. REQUIREMENTS

3.1 Standard sample - The cloth shall match the standard sample for shade and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3).

3.1.1 Formula approval - The prospective supplier or his finishing plant shall submit with his bid to the contracting officer for approval, a list of materials which will be used if the contract is awarded. The list shall stipulate amounts, grades, and standards of all materials including the solvents, and the manufacturer's name. This list is for the information of the Government and is not to be disseminated to the public. Approval of the quantitative formula submitted shall in no way absolve the supplier from complying with the performance requirements of this specification.

3.2 Material - The cloth shall be made from cotton which has been carded and spun into singles yarn for both the warp and filling.

3.3 Weave - The weave shall be a 5-harness sateen. The weave of the cloth when viewed from the filling-effect side shall appear as shown on Figure 1. The filling-effect side shall be finished and identified as the "face" side (see 3.9).

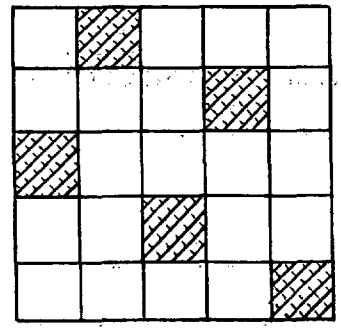


Figure 1

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3.4 Physical requirements - The finished cloth shall conform to the requirements of Table I when tested as specified in 4.4.

TABLE I

PHYSICAL REQUIREMENTS

Characteristic	Requirement
Weight, ounces per square yard, minimum	4.0
Yarns per inch, minimum	
Warp	100
Filling	150
Breaking strength, pounds, minimum	
Warp	50
Filling	60
Shrinkage in laundering, percent, maximum	
Warp	2
Filling	2
Acidity, pH	5.0-8.5

3.5 Finish - The cloth shall be singed, mercerized, dyed, and fire retardent treated. The preshrinking process used shall not be identified on the cloth, the ticket, or on the package.

3.5.1 Fire retardancy - The treated cloth shall show no more than 2.0 seconds unit average after flame time, not more than 2.0 seconds unit average after glow time, not more than 5.5 unit average char length when tested as specified in 4.4.

3.6 Color - Unless otherwise specified, the color of the finished cloth shall be as specified by the procuring activity. The use of dyes and compounds containing elementary sulfur capable of oxidation to sulfuric acid is prohibited. The dyestuff shall be chosen and applied so that the dyed and finished cloth shall contain no more labile sulfur than shown by the standard sample when tested as specified in 4.4. When a standard sample is not available, the dyed and finished cloth shall show no more than a slight trace of labile sulfur as defined in the test method when tested as specified in 4.4.

3.6.1 Matching - The dyed and finished cloth shall match the standard sample under natural (north sky) daylight or artificial daylight having a color temperature of 7500° Kelvin and shall be a good approximation to the standard sample under incandescent lamplight at 2800° Kelvin.

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3.6.2 Colorfastness - The dyed and finished cloth shall show fastness to wet crocking, laundering, and light equal to or better than the standard sample. When no standard sample is available, the cloth shall show a minimum of "good" fastness to wet crocking, laundering, and light.

3.7 Nonfibrous material - The total sizing, finishing, and other nonfibrous material content of the cloth prior to treatment shall not exceed 8.5 percent when tested as specified in 4.4.

3.8 Fiber identification - Each piece shall be labeled, ticketed, or invoiced for fiber in accordance with the Rules and Regulations Under the Textile Fiber Products Identification Act.

3.9 Face identification - The filling-effect side of the cloth shall be identified as the face by stamping the word "Face" on that side at each end of the piece.

3.10 Width - The width shall be specified and shall be not less than the minimum acceptable width inclusive of selvages.

3.11 Length - Unless otherwise specified, the cloth shall be furnished in continuous lengths, each not less than 40 yards. Each length shall be put-up in a roll as specified in PPP-P-1134.

3.12 Workmanship - The quality of manufacture shall be controlled to guarantee conformity to the high standards of workmanship required by the acceptable quality levels specified in Section 4.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Classification of inspection - The inspection requirement specified herein shall be classified as quality conformance inspection (see 4.2).

4.1.2 Certificate of compliance - Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

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4.2 Quality conformance inspection - The quality conformance inspection shall consist of the examination and tests of all the requirements specified herein.

4.2.1 Component and material inspection - In accordance with 4.1 above, components and materials shall be tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable procurement documents.

4.2.2 Examination of the end item - Examination of the end item shall be in accordance with 4.2.2.1 through 4.2.4. and MIL-STD-105.

4.2.2.1 Yard-by-yard examination - The entire yardage of each piece in the sample, but not more than 100 yards per piece, shall be examined on the face side. Defects as defined in Section I of FED-STD-4 which are clearly noticeable at normal inspection distance (3 feet) shall be scored and assigned demerit points as listed in 4.2.1.1, except that only those slubs and knots which exceed the maximum limit shown on Figure 1 shall be scored. No linear yard from any one piece within the sample shall be penalized more than 4 points. The sample size shall be twenty pieces selected from twenty containers. For determination of acceptability on the average quality of the lot, the lot shall be unacceptable if the points per 100 square yards exceeds 25.0 points. For determination of acceptability on the individual piece quality basis, the lot shall be unacceptable if two or more individual pieces exceeds 38.0 points. If no individual piece exceeds the point level, the lot shall be accepted. If one piece exceeds the point level, a second sample of 20 pieces shall be selected for individual piece quality examination only. If no piece in the second sample exceeds the point value, the lot shall be accepted. If one or more pieces exceeds the value, the lot shall be rejected. Point computation for lot quality and individual piece quality shall be as follows:

$$\frac{\text{Total points scored in sample size X 3600}}{\text{Contracted width of cloth (inches) X Total yards inspected}} = \text{Points per 100 square yards}$$

4.2.2.1.1 Demerit points - Demerit points shall be assigned as follows:

For defects 3 inches or less in any dimension	- one point
For defects exceeding 3 inches but not exceeding 6 inches in any dimension	- two points
For defects exceeding 6 inches but not exceeding 9 inches in any dimension	- three points
For defects exceeding 9 inches in any dimension	- four points

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The following defects, when present, shall be scored four points for each yard in which they occur: Baggy, ridgy or wavy cloth

Selvage tight, slack, loopy, wavy, scalloped, curled, folded rolled, cut and broken.

Objectionable odor.

Width less than specified.

Poor dye penetration, mottled, streaky or cloudy.

4.2.2.1.2 Limits of slubs and knots - Only slubs and knots which exceed the limits shown on Figure 1 of FED STD-4 shall be scored.

4.2.2.2 Examination for length -

4.2.2.2.1 Individual pieces - During the yard-by-yard examination, each piece in the sample shall be examined for length. Any length found to be less than the minimum specified or more than two yards less than the length marked on the ticket shall be considered a defect with respect to length. The lot shall be unacceptable if two or more pieces in the sample are defective in respect to length.

4.2.2.2.2 Total yardage in sample - The lot shall be unacceptable if the total of the actual lengths of pieces in the sample is less than the total of the lengths marked on the tickets.

4.2.3 Examination for shade - During the yard-by-yard examination, each piece in the sample shall be examined for shade. Any piece in the sample off shade, shaded side to side, side to center, or end to end shall be cause for rejection of the entire lot represented by the sample.

4.2.4 Examination for face identification, non-identification of preshrinkage process, and compliance with Textile Fiber Products Identification Act - During the yard-by-yard examination, each piece in the sample shall be examined for defects listed below. The lot shall be unacceptable if two or more of the following defects are present in the sample:

Face identification missing from either or both ends

Face identification on wrong side

Preshrinkage process identified by name or trademark on cloth or ticket

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Not labeled or ticketed in accordance with the Rules and Regulations Under the Textile Fiber Products Identification Act

4.3 Examination of preparation for delivery requirements - An examination shall be made in accordance with the provisions of PPP-P-1134 to determine that packaging, packing and marking requirements of Section 5 of this specification are complied with.

4.4 Testing of the end item - The methods of testing specified in FED-STD-191, wherever applicable, and as listed in Table II shall be followed. The physical and chemical values specified in Section 3 apply to the results of the determinations made on a sample unit for test purposes as specified in the applicable test method. The sample unit shall be 3½ continuous yards, full width of the finished cloth and 1/4 yard prepared cloth before treatment. The sample size (number of sample units) shall be as shown below. The lot size shall be expressed in units of 1 yard. The lot shall be unacceptable if one or more sample units fail to meet any test requirement specified. All test reports shall contain the individual values utilized in expressing the final results.

Lot size (yards)	Sample size
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

TABLE II

TEST METHODS

Characteristic	Requirement paragraph	Test Method
Weave	3.3	Visual 1/
Weight	3.4	5041
Yarns per inch		
Warp	3.4	5050
Filling	3.4	5050
Breaking strength		
Warp	3.4	5100
Filling	3.4	5100

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TABLE II (Continued)

Characteristic	Requirement paragraph	Test method
Colorfastness to		
Light	3.6.2	5660
Laundering	3.6.2	5610
Crocking	3.6.2	5651
Shrinkage	3.4	5550
Flame resistance		
Initial		
After flame time, sec.	3.5.1	5903
After glow time, sec.	3.5.1	5903
Char length, inch	3.5.1	5903
Presence of labile sulfur	3.6	2020
pH	3.4	2811
Nonfibrous material	3.7	2611

1/ One determination for each sample unit and the result reported as "pass" or "fail".

5. PREPARATION FOR DELIVERY

5.1 Packaging - Packaging shall be Level A or C as specified (see 6.2 and 6.2.1).

5.1.1 Level A or C - The cloth, put up as specified, shall be packaged in accordance with the applicable requirements of PPP-P-1134.

5.2 Packing - Packing shall be Level A, B, or C as specified (see 6.2).

5.2.1 Levels A, B, and C - The cloth shall be packed in accordance with the applicable requirements of PPP-P-1134.

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5.3 Marking - In addition to any special marking required by the contract or order, shipments shall be marked in accordance with the applicable requirements of PPP-P-1134.

6. NOTES

6.1 Intended use - The cloth covered by this specification is intended for use as black-out curtains in aircraft.

6.2 Ordering data - Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Color.

(c) Width required.

(d) Selection of applicable levels of packaging and packing.

6.2.1 In the preparation of contracts or orders, it must be noted that various options, choices, and alternatives, as indicated in PPP-P-1134, may be exercised in the preparation for delivery of this cloth.

6.3 Standard sample - For access to standard sample, address the procuring office issuing the invitations for bids.

6.4 Approval of flame resistant treatments for utilization under this specifications is the responsibility of the contracting officer and is based on extensive tests, including those for toxicity, which are not set forth in this specification. Because of the time necessary to conduct full evaluations, only those treatments approved and so listed in the invitation for bids or request for proposals shall be considered acceptable for the related procurement. Information on currently approved material may be obtained from Defense Personnel Support Center, Philadelphia, Pennsylvania 19101.

6.5 This specification supersedes Navy Purchase Description NCTO/TD-PD-107 dated 10 September 1962.

Preparing Activity:
Navy - AS
Project No. 8305-N063

☆ U. S. GOVERNMENT PRINTING OFFICE: 1971-433-696/5409

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

MIL-C- 81818 (AS) CLOTH, SATEEN, COTTON, FIRE-RETARDANT TREATED

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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